Hardness Evidence Lab

BUILD · Science · 40 minutes

I can use the same stated scratch method to compare four coded rock samples and give a cautious evidence-based conclusion.

Prepare before pupils arrive

Setup: Pretest and code all samples, lock the method and prepare one complete fixed-data route. Keep a visible stop/check box beside every code and model preserving an unexpected result.

Safety: Teacher or trained adult controls the secured tool. Remove unstable samples, require assessed eye protection, contain debris and stop rather than increase force. The no-tool parity route carries the same science goal.

Equipment: Four teacher-pretested coded dry samples; stable individual trays; secured blunt scratch jig or teacher demonstration; fixed guide; wipes; hand lens; four-row result table; eye protection where assessed.

Safety: Follow the local risk assessment. No hammering, breaking, tasting, smelling or dust blowing. An adult controls the secured tool. Stop on flaking, crumbling or dust; preserve the note and switch to fixed-data parity. Wash hands.

Fallback: Direct the adult through the method or use the printed fixed observations. Recorder, observer and conclusion roles are complete participation routes.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Rocks: test hardness.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Define relative hardness using scratch resistance. Name three controls in the stated method.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	relative — compared with the others repeat — same method again conclusion — what the results support Lock the method before Code A: same secured tool, dry start, guide length, passes, safe force proxy, wipe and viewing rule. Predict, run or reveal, wipe and record one code at a time.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Order only the tested codes by the remaining groove evidence: deeper groove means lower resistance to this scratch method. Use a cautious conclusion such as ‘Code D resisted a groove more than Code A under this method.’ Add one limitation: specimens vary, the force proxy is simple and one classroom tool does not produce a universal hardness number.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each after-wipe result, method card and limitation in its correct evidence role. Lock the method.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Run or direct the same safe scratch method on four coded samples, record every after-wipe result, repeat one check where appropriate and write a cautious conclusion. Record: A complete four-code raw-results table, one preserved repeat/check note, a relative comparison using at least two codes and one method or sample limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Use two code results to give a relative-hardness conclusion and state one limit. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Keep the faint-groove result and, if useful, repeat the same method.

Correct. A valid unexpected result remains evidence; a stated repeat can check how dependable it is.

Misconception repair

Repaired. Scientific records keep all valid raw observations; a summary can be added but cannot replace them.

Guided checkpoint

Correct. A valid unexpected result remains evidence; a stated repeat can check how dependable it is. Predictions do not control observations. Preserve what happened under the stated method.

Code A

DEEP GROOVE. Code A showed the lowest scratch resistance in this four-code model set. Use the depth that remains after wiping.

Code B

SHALLOW GROOVE. Code B records a shallow groove; the safety note remains beside the result. Keep the result and the stop/check note together.

Code C

FAINT GROOVE. Code C records a faint groove, even if the prediction was no groove. Do not promote a faint remaining indentation to no groove.

Code D

NO GROOVE AFTER WIPE. Code D resisted a groove most in this representative four-code set. The result is based on the after-wipe surface.

Keep first result, then repeat

METHOD / REPEAT RULE. Method rule: repeats add evidence without erasing the first observation. This card controls evidence handling rather than ranking a groove.

These samples, this tool, this method

LIMIT OF THE CONCLUSION. Limit: the conclusion belongs to the tested codes and stated method. This card restricts how far the conclusion can travel.

Model limitation

The fixed data represent particular teaching samples and one scratch method. They do not assign Mohs values, measure toughness or approve a rock for construction.

Independent evidence

A complete four-code raw-results table, one preserved repeat/check note, a relative comparison using at least two codes and one method or sample limitation.

Supported: Match four after-wipe picture cards to Codes A–D and complete one two-code comparison. Use deep, shallow, faint and none symbols. Adult handling and exact scribing are valid.

Standard: Complete all four rows, keep one repeat/check note and write a conclusion comparing at least two codes. Use: Code __ showed __ while Code __ showed __, so under this method __.

Stretch: Evaluate the relative pattern, explain an unexpected result and propose one safe improvement without claiming a universal hardness value. Separate sample variation, method variation and properties the test did not measure.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre’s authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner’s practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W10B_Hardness_Evidence_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.